

Karan Shah

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Education

SUNY, Stony Brook University

MS IN COMPUTER SCIENCE | GPA : 3.64/4

Coursework: Big Data Analytics, Theory of Database Systems, Machine Learning, Computer Vision, Data Science, Probability and Statistics

Stony Brook, NY

Aug. 2019 - **Present**

Gujarat Technological University

B.E. IN COMPUTER ENGINEERING

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Ahmedabad, India

Aug. 2014 - May. 2018

Skills

Programming Python, Javascript, JAVA, C/C++, Golang

Tools/Technologies AWS, MapReduce, Spark, Hadoop, Docker, Jenkins, Kubernetes, Git, SQL, NoSQL, Linux, Matlab

Frameworks/Libraries NodeJS, Django, Flask, Angular, ElectronJS, Scikit-Learn, Numpy, Pandas, PyTorch, Tensorflow, Keras, OpenCV

Work Experience

SUNY, Stony Brook University

GRADUATE TEACHING ASSISTANT

• Assisted the professor in Course CSE327-01: **Computer Vision** for assignments formulations, assessments and resolving queries of students

Stony Brook, NY

January 2020 - May 2020

Simform Solutions

SOFTWARE ENGINEER

- Developed and automated **ETL** jobs to work on a huge realtime data using AWS services (Glue, Lambda, S3) and Spark
- Developed an end to end **Video Streaming system**; able to upload, scan and host the content on the cloud, and stream on cross-platform desktop application over globally separated clients; using **AWS services (Lambda, S3, Cloudfront, MQTT), NodeJS, and Python**
- Developed software system to semi-automate Company's marketing work with Chrome Extension and **REST API** Web backend using **Python and NodeJS**, efficient in reducing ~ **40 %** of Marketing team's manual work

Ahmedabad, India

June 2018 - April 2019

ISRO - Indian Space Research Organization

RESEARCH INTERN, MACHINE LEARNING

- Designed generalized **Convolutional Neural Network** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for data preparation/processing of geographic remote sensing data, visualization, real-time training performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Ahmedabad, India

July 2017 - May 2018

Publication

Band-wise Independent Pansharpening Using Convolutional Neural Networks with Shared Weights

PANKAJ BODANI, KARAN SHAH, SHASHIKANT SHARMA

- Submitted to **IEEE** Geoscience and Remote Sensing Letters

Projects

Child Mortality Analysis, Under Supervision of Prof. Andrew Schwartz (Python, Dask, Spark)

April 2020 - May 2020

- Analysed secondary factors contributing to child mortality through Big Data Techniques
- Implemented **Linear Regression and Similarity Search** running on women lifestyle real data distributed on Hadoop clusters

Retail Sales Data Analysis, Under Supervision of Prof. Steven Skiena (Python, Pandas)

Oct. 2019 - Dec. 2019

- Worked on a market basket dataset of a local chain retail stores to provide insights to help the company with their business
- Analysed data to find reasons for drop in sales with additional help of external datasets
- Implemented **Word2Vec and LSTM model with k-means clustering** to improve Shelf Space Management and Inventory management

Video Action Classification and Recognition (Python, PyTorch)

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (inclass 3rd rank) on image test data, and **84%** (inclass 4th rank) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

Company's Internal Communication System (Java, Javascript)

Sept. 2018 - Dec. 2018

- Created a web-based software for employees of a Software Company to send/receive messages, share files in person, groups and broadcast channels with various administrative permission roles
- Utilised **SQLite** to store data of each user with **Java** Servlets to query the database and applied Software Engineering approaches

Web Search Engine (Java, Php, Python, Ajax, Apache Solr, Apache Tika)

Jan. 2018 - April. 2018

- Developed a search engine using **Java, PHP, AJAX** which implements Apache Solr PageRank and Lucene algorithms
- Crawled web and gathered data using Crawler4j and indexed the webpages using Apache Tika parser
- Added **advanced features** like spell checking, autocompletion and generation of snippets for improving result display

Desktop Application to Synchronize Local Storage with Cloud (AWS, Node.js, Electron.js)

March 2017 - Apr. 2017

- Built cross-platform application leveraging the ElectronJS framework
- Used Node.js and AWS services to fully synchronize local storage and AWS S3 storage